
UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

Anonymous authors
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ABSTRACT

Vision-language models (VLMs) now report strong scores across visual question answering benchmarks that cover visual reasoning, document understanding, and multimodal dialogue, but the same aggregate progress does not fully characterize their fine-grained visual knowledge. We evaluate a broad set of recent VLMs on traditional fine-grained classification benchmarks and frame the results against both general VLM scores and the underlying vision-encoder baselines. This comparison reveals clear empirical separation: models with similar general VQA averages can differ substantially on fine-grained accuracy, and most VLMs remain below their corresponding CLIP-style encoders. To explain these margins, we run 22 controlled ablations over base LLMs, vision encoders, pretraining data, and training strategy. Stronger LLMs improve fine-grained and general VQA scores by comparable amounts, whereas stronger vision encoders and pretraining choices yield more targeted gains for fine-grained classification. In particular, pretraining is a consistent driver of fine-grained performance, especially when the language model is unfrozen rather than leaving only the connector trainable. Together, these results identify which existing design choices most reliably close fine-grained accuracy gaps and provide practical evidence for building more vision-centric VLMs.

1 INTRODUCTION

Vision-language models (VLMs), which combine vision encoders with large language models (LLMs), have achieved increasingly strong benchmark results on visual interpretation and reasoning tasks (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b). Their reported gains span open-ended visual question answering, document understanding, and multimodal dialogue (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). These aggregate scores, however, leave an important quantitative question unresolved: when evaluation requires distinguishing visually similar categories, how much fine-grained visual knowledge do current VLMs retain, and which design choices most improve that accuracy?

Fine-grained visual recognition—the ability to distinguish between visually similar categories—tests whether a model can use subtle visual cues while remaining robust to intra-class variation in pose, lighting, and background. Traditional vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have been measured extensively on these classification benchmarks, providing strong quantitative reference points. Newer VLMs are typically compared on broader VQA-style metrics instead, so their fine-grained classification accuracy and margins relative to encoder-only baselines remain less clearly characterized.

Measuring this capability matters because fine-grained perception is often the prerequisite for downstream reasoning. Many practical VLM deployments involve long-tailed categories where the decision boundary is between closely related options rather than between coarse object types. For example, if a model misidentifies a mushroom species, its textual knowledge about toxicity cannot support a reliable safety judgment. Quantifying which components improve these distinctions therefore gives direct evidence for designing architectures and training recipes that preserve visual detail.

We begin with a comparative evaluation that places fine-grained classification accuracy next to general VLM benchmark performance. We test 15 VLMs, including LLaVA (Liu et al., 2023; 2024a),

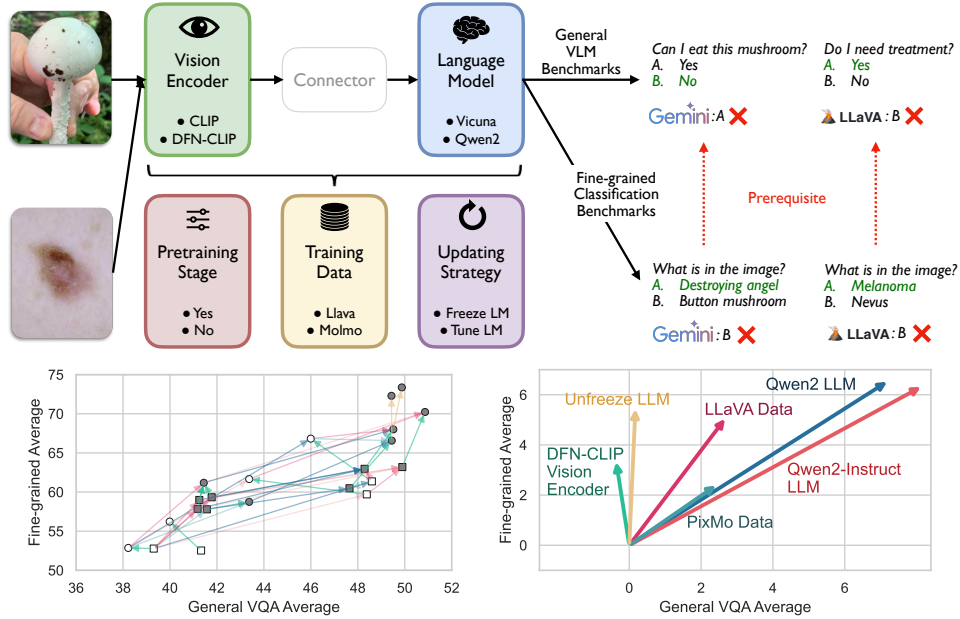


Figure 1: **Overview.** (Top) We measure VLM fine-grained classification accuracy as a vision-centric capability that is not summarized by standard VQA averages. (Bottom) Across 22 controlled ablations of model components and training strategies (left), the comparison pattern isolates which changes produce the largest and most consistent fine-grained gains (right).

Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al., 2024), on four fine-grained classification benchmarks: ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014). We then compare those accuracies with aggregated results from eight general VLM benchmarks, including MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). The resulting pattern is not a single monotonic ranking: models with comparable general scores can have markedly different fine-grained accuracies, and VLMs often remain below their CLIP-based vision encoders, consistent with prior observations (Zhang et al., 2024; Geigle et al., 2024).

To identify which factors account for these margins, we conduct a systematic ablation study over base model choices and training paradigms. The comparisons isolate the effects of the LLM, the vision encoder, pretraining, data quality, and which weights are updated during training. Across 22 controlled ablations, the quantitative trends separate changes that broadly lift all metrics from changes that disproportionately improve fine-grained classification:

- **Language model (§4.2):** Upgrading the base language model raises both fine-grained classification and general VLM benchmark scores by similar margins.
- **Vision encoder (§4.2):** Replacing the vision encoder with a stronger alternative yields larger gains on fine-grained classification than on general VLM benchmarks, especially after two-stage training.
- **Pretraining stage (§4.3):** Image-caption pretraining consistently improves fine-grained scores, with smaller corresponding changes on general VLM benchmarks.
- **Weight updating method (§4.3):** Training both the connector and the LLM during pretraining improves fine-grained accuracy relative to connector-only pretraining.
- **Data quality (§4.3):** Differences in pretraining and instruction-tuning data quality produce comparatively small changes, particularly when language model weights remain frozen during pretraining.

Overall, the evidence positions fine-grained visual perception as a measurable VLM capability that current benchmark averages can obscure. By comparing existing models, their vision encoders, and controlled ablation variants under the same evaluation framing, we identify which design choices most consistently improve fine-grained knowledge. These results provide practical guidance for developing vision-centric VLMs intended for long-tail settings where small accuracy margins on visually similar categories can matter substantially.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) integrate visual inputs with language models, often using the LLaVA-style architecture (Liu et al., 2023) with three measurable components: a vision encoder, a language model, and an MLP connector between them. Recent VLMs report strong results on multimodal tasks such as visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). The corresponding evaluation ecosystem includes benchmarks such as MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021), which provide useful aggregate comparisons across models. These benchmarks, however, primarily score reasoning and language understanding conditioned on visual inputs, leaving object recognition and fine-grained classification less directly measured. Our work fills this empirical gap by evaluating VLMs on fine-grained classification and by comparing those scores to the general benchmark averages that usually define model progress.

Fine-grained visual classification. Fine-grained classification is a long-standing computer vision evaluation setting in which the competing labels are visually similar subcategories, such as bird species (Wah et al., 2011), flowers (Nilsback & Zisserman, 2008), and pet breeds (Parkhi et al., 2012). Because these datasets expose small visual margins between classes, they have served as important quantitative testbeds for vision encoders including ResNet (He et al., 2016), CLIP (Radford et al., 2021), and DINO (Caron et al., 2021). Recent VLM evaluations have shifted attention toward VQA, so the same classification benchmarks are less often used to report VLM capability. This omission matters for practical settings with long-tailed categories and precise recognition requirements, including medical diagnosis, food safety, and species identification. Evaluating VLMs on fine-grained benchmarks therefore complements language-centric reasoning scores with a direct measure of vision-centric usability.

Evaluation and ablation of VLMs for fine-grained classification. Recent studies have begun to measure VLMs on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024), but the reported comparisons are typically limited to a small number of models or datasets. We extend this evidence base by evaluating a broader set of VLMs across multiple fine-grained classification benchmarks under a unified multiple-choice protocol. We also add controlled ablations that make the comparison structure explicit: each experiment changes a model component or training choice and reports how the margin shifts for fine-grained classification relative to general VQA.

Other recent work studies the VLM design space through architecture choices (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and multimodal dataset composition (Gadre et al., 2023; Udandarao et al., 2024; Deitke et al., 2024). Those analyses are usually optimized around general benchmarks that emphasize language and reasoning. In contrast, we quantify how base model selection, pretraining method, and pretraining or instruction-tuning data composition affect fine-grained classification accuracy, making the practical significance of each design choice visible for vision-centric VLM development.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

This section defines the fine-grained evaluation axis, describes the benchmark conversion and testing protocol, and reports the comparison patterns observed across 15 existing VLMs.

3.1 MOTIVATION

Fine-grained classification measures whether a model can distinguish visually similar subcategories within a broader category, such as bird species or car models (Wei et al., 2021). Benchmarks such as ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008) have therefore become standard quantitative testbeds for training and evaluating vision encoders. With large-scale self-supervised and supervised learning, encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) reached near- or superhuman accuracy on challenging fine-grained classification tasks, establishing strong baselines for comparison.

These same classification benchmarks are less prominent in current VLM evaluation. Models such as LLaVA (Liu et al., 2023) are usually compared on visual question answering (VQA), including

academic VQA benchmarks such as MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or document understanding tasks such as DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). Those metrics are valuable for reasoning and language processing, but they contain fewer direct tests of fine-grained object categories, leaving an important vision-centric score underreported.

Fine-grained classification is practically important because real-world VLM inputs often fall in long-tailed distributions where the relevant alternatives are visually close. If a blind user asks whether a mushroom is safe to eat, the model must first identify the species before textual knowledge about toxicity can be useful; confusing a deadly destroying angel with an edible white button mushroom would invalidate the downstream answer. The same comparison structure appears in medical diagnosis, where similar visual presentations can imply different treatments, and in driving, where red signs with related shapes can require different actions. High fine-grained accuracy is therefore not only a benchmark preference but also a reliability requirement for applications that depend on precise visual distinctions.

3.2 BENCHMARKS

We measure VLM fine-grained visual knowledge with four established object-recognition benchmarks that cover different domains and class granularities: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014).

The datasets contain 37 to 1,000 classes, so directly presenting every class as an answer option would not match common VLM multiple-choice evaluation formats. Following Geigle et al. (2024), we convert each dataset into 5-way multiple-choice questions while retaining difficult distractors. For each image, an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) selects the four hardest incorrect labels by image-label cosine similarity, and these negatives are paired with the ground-truth label. We use OpenCLIP for negative mining to avoid favoring or penalizing VLMs built on OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023), each of which uses different training data. This protocol keeps the evaluation in a format VLMs are designed to answer while preserving much of the original classification difficulty. For example, CLIP ViT-L/14-336px (Radford et al., 2021) scores 75.2% on the original 102-class Flowers-102 task and 78.3% on our 5-way version, indicating that the multiple-choice conversion does not make the task trivially easy.

3.3 TESTING SETUP

We evaluate 15 recent VLMs in the 7B–13B parameter range on the converted 5-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abdin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b). For each VLM, we use its default VLMEvalKit prompt format (Duan et al., 2024) and compute accuracy by exact match against the multiple-choice options. For encoder-only baselines, we evaluate CLIP-family models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) using standard zero-shot classification: images and class-label prompts (“A photo of a [class]”) are encoded by the image and text towers, and the top-1 label is selected by cosine similarity.

To test whether standard VLM metrics already capture fine-grained ability, we compare each model’s average accuracy over the four fine-grained benchmarks with its VLMEvalKit average over eight general VQA benchmarks, including MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024).

3.4 RESULTS

Figure 2 plots each tested VLM by two quantities: mean fine-grained classification accuracy across our four benchmarks and mean general VQA score across eight existing benchmarks. The scatter shows that similar general VQA performance does not imply similar fine-grained performance. For example, CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a) both score near 48% on the general VQA average, yet their fine-grained accuracies differ by 19pp (percentage points): 77.3% for CogVLM-Chat versus 58.1% for LLaVA-NeXT-Vicuna-13B. This margin makes fine-grained classification a distinct empirical axis rather than a redundant proxy for existing VLM benchmark averages.

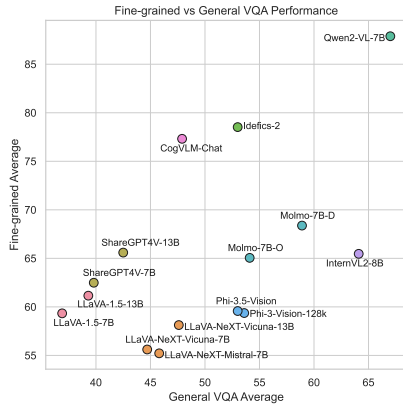


Figure 2: **Fine-grained classification versus general VQA across VLMs.** Models with similar eight-benchmark VQA averages can differ substantially in four-benchmark fine-grained accuracy, showing that the two metrics capture different empirical patterns.

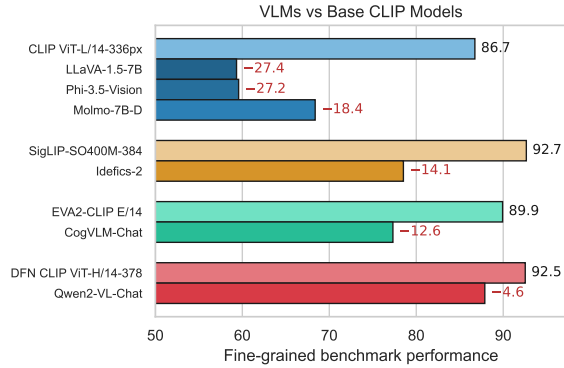


Figure 3: **VLMs compared with their corresponding CLIP vision encoders.** The encoder baseline provides an upper reference for retained fine-grained knowledge: Qwen2-VL-Chat is closest to its DFN-CLIP encoder, while the other tested VLMs show larger accuracy gaps.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

Because current VLMs are built on pretrained vision encoders, the multiple-choice protocol also enables a direct comparison between each VLM and the corresponding CLIP-style encoder on the same fine-grained tasks. Figure 3 shows that most VLMs do not preserve the full fine-grained accuracy of their encoder baselines. Molmo (Deitke et al., 2024), despite strong VQA performance, reaches 68.4% average accuracy in our multiple-choice setting, compared with 86.7% for its CLIP ViT-L/14 vision encoder (Radford et al., 2021). The strongest tested VLM, Qwen2-VL-Chat (Wang et al., 2024b), narrows the gap but still trails its DFN-CLIP (Fang et al., 2023) encoder by 4.6 points. These encoder-to-VLM margins motivate the controlled ablations in the next section, where we identify which design choices most consistently recover fine-grained classification performance.

Finding 2: A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

Building on the model-level comparisons in Section 3, we run controlled ablations to determine which design differences account for fine-grained accuracy margins. The experiments focus on two comparison families: base model selection and training methodology. Unless otherwise stated, we use the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023), train for one pretraining epoch when pretraining is included, and then run one finetuning epoch. We keep LLaVA-1.5’s default hyperparameters, including learning rate and batch size, fixed across experiments so that reported differences can be attributed to the ablated factor.

We report fine-grained performance as the average score over the four classification benchmarks from Section 3.2. For general VQA performance, we report the average score over three widely used multiple-choice benchmarks: MMMU (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b). These three benchmarks provide a compact general VQA comparison set with broad domain coverage and widespread use in VLM evaluation.

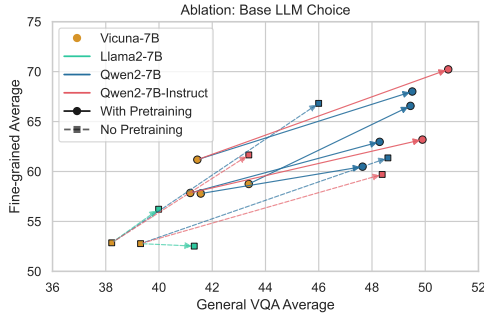


Figure 4: **Base LLM ablation.** Replacing Vicuna-7B in the LLaVA setup with Qwen2-7B produces nearly matched gains on both metrics: +7.5pp for fine-grained classification and +7.7pp for general VQA.

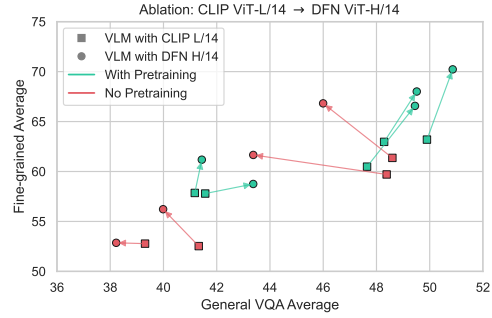


Figure 5: **Vision encoder ablation.** Replacing CLIP ViT-L/14 with DFN-CLIP ViT-H/14 yields a larger fine-grained gain (+4.5pp) than general VQA gain (+1.2pp) when the encoder is sufficiently integrated through pretraining.

4.2 BASE MODELS

The evaluation in Section 3 shows that recent VLMs often differ in their base LLMs, while many still rely on the CLIP ViT-L/14 vision encoder from Radford et al. (2021). We therefore isolate the contribution of each base model choice and compare its effect on fine-grained accuracy against its effect on general VQA.

4.2.1 LLM CHOICE

Setup. Starting from the LLaVA-1.5-7B setup (Liu et al., 2023), we vary only the LLM while evaluating the change across different vision encoders and pretraining configurations. The comparison includes Vicuna-7B-1.5 (Chiang et al., 2023), the original LLaVA LLM; Qwen2-7B (Yang et al., 2024), used in Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024); Llama-2-7B (Touvron et al., 2023), the base model for Vicuna; and Qwen2-7B-Instruct, the instruction-tuned Qwen2-7B variant.

Findings. Figure 4 shows a broad, metric-balanced effect from replacing Vicuna-7B with Qwen2-7B: the average gain is +7.5pp (percentage points) on fine-grained benchmarks and +7.4pp on general VQA benchmarks. Qwen2-7B-Instruct follows the same pattern, with +7.5pp and +8.1pp gains, respectively. Llama-2-7B, the non-instruction-tuned base of Vicuna, produces smaller improvements, although the practical implications of using a non-instruction-tuned LLM remain unclear. All trained models, including the lower-scoring Vicuna-based variants, consistently output valid multiple-choice responses. Thus, the fine-grained gains are better explained by improved option knowledge or reasoning over the visual evidence than by improved answer-format adherence.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

4.2.2 VISION ENCODER CHOICE

Setup. Most evaluated VLMs, including the LLaVA series and Molmo, use OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) as the vision encoder. Qwen2-VL (Wang et al., 2024b), by contrast, uses DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which has stronger zero-shot object-recognition performance on tasks such as ImageNet. We compare these two encoders while varying LLM choices and pretraining configurations, allowing us to measure whether a stronger encoder transfers more to fine-grained classification than to general VQA.

Findings. Figure 5 shows that the DFN-CLIP substitution has different effects depending on whether connector pretraining occurs before finetuning. Without pretraining, DFN-CLIP improves fine-grained scores by +2.8pp but reduces general VQA by -2.5pp. After connector pretraining, the pattern becomes more favorable: general VQA is stable to slightly higher (+1.2pp), while fine-

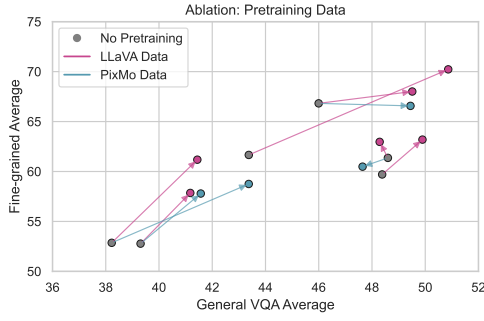


Figure 6: **Pretraining data ablation.** Adding LLaVA (CC-3M) or Molmo (PixMo-Cap) pretraining data improves both metric groups with similar patterns: LLaVA gives +4.0pp fine-grained and +2.1pp general gains, while PixMo gives +2.4pp and +2.5pp.

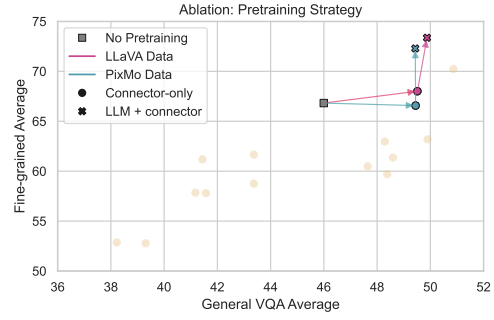


Figure 7: **Pretraining update ablation.** Compared with connector-only pretraining, unfreezing the LLM during pretraining adds +5.5pp on fine-grained classification while preserving general VQA scores.

Table 1: **Key ablation settings and comparison to existing models (bolded).** The table separates fine-grained and general VQA effects: upgrading the LLM improves both metrics, stronger vision encoders and LLM-unfrozen pretraining contribute more to fine-grained accuracy, and instruction finetuning produces a smaller measured change.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

grained performance rises by +4.5pp. The larger fine-grained margin indicates that stronger vision encoders provide targeted benefits once the VLM has been trained to integrate their representations.

Takeaway 2: Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pretraining before the finetuning stage.

4.3 TRAINING METHODS

Recent VLMs are typically trained in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). The pretraining stage uses large-scale image-caption data to align the vision encoder with the LLM, while instruction tuning uses curated multi-round multimodal instruction data to improve task following and human alignment. We evaluate these stages by measuring how each training choice changes fine-grained classification accuracy relative to general VQA performance.

4.3.1 PRETRAINING

Setup. We structure the pretraining ablations around three quantitative questions:

- How much does pretraining add for LLaVA-architecture models whose base components are already pretrained on billion-scale data?
- How do the gains from connector-only pretraining compare with the gains from also unfreezing the LLM?

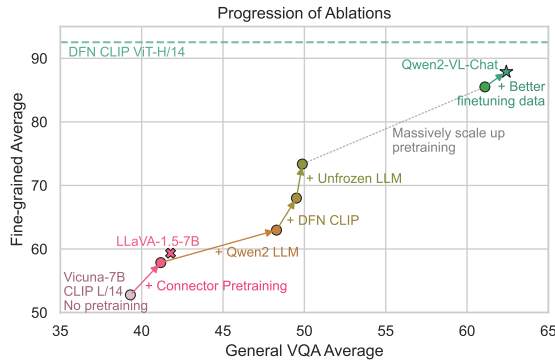


Figure 8: **Progression of combined ablation effects.** Starting from Vicuna-7B with CLIP ViT-L/14, successive changes to the LLM, vision encoder, and training setup raise fine-grained accuracy from 52.8% to 73.4%. This closes more than half of the gap to Qwen2-VL-Chat’s 87.9%, with the remaining difference primarily attributed to Qwen2-VL’s much larger 1.4T-token pretraining scale.

- How sensitive are downstream metrics to pretraining data quality?

We compare three pretraining data conditions: LLaVA pretraining data (Liu et al., 2023), a CC-3M subset with web-scraped captions; Molmo pretraining data (Deitke et al., 2024), PixMo-Cap with detailed human-annotated captions; and no pretraining. For most runs, only the connector is pretrained, matching Liu et al. (2023). We also test a Deitke et al. (2024)-style update schedule in which the connector is trained alone for the first 20% of steps and the LLM is unfrozen for the remaining 80%. Every configuration is then finetuned for one epoch on LLaVA finetuning data, keeping the downstream comparison fixed.

Findings. Figure 6 summarizes the pretraining data ablation. Connector-only pretraining improves most VLM configurations, with the Qwen2 and CLIP ViT-L/14 pairing as the main exception. The gains are especially large for Vicuna-based VLMs: fine-grained classification increases by +6.1pp on average, while general VQA increases by +3.1pp. This larger fine-grained margin indicates that pretraining is particularly important for recovering detailed visual distinctions.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

The comparison between LLaVA web-scraped captions and PixMo human-annotated captions shows relatively small differences under this setup. Relative to LLaVA data, PixMo changes fine-grained performance by -1.6pp and general VQA by +0.4pp. Thus, in connector-only pretraining, caption quality differences are not the dominant source of the observed accuracy shifts.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance.

One possible explanation is that connector-only pretraining limits how much the model can benefit from richer captions. We therefore pretrain both the LLM and connector after a 20% connector-only warmup. Figure 7 shows that this update strategy adds +5.5pp to fine-grained classification without reducing general VQA scores. Because the improvement appears with both LLaVA and PixMo pretraining data, the evidence suggests that unfreezing the LLM during pretraining matters more than the caption-quality contrast in these experiments.

Takeaway 5: Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

4.3.2 FINETUNING

Setup. For the finetuning comparison, we start from the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b), the base model used to train Qwen2-VL-7B-Chat, and finetune it on LLaVA-Instruct data (Liu et al., 2023) using the same protocol as above. We compare this model to

Qwen2-VL-7B-Chat, which uses 2M in-house instruction-finetuning examples (Wang et al., 2024b). This isolates the effect of the instruction-finetuning dataset while holding the pretrained base model family fixed.

Findings. Finetuning Qwen2-VL-7B-Base only on LLaVA finetuning data slightly lowers both metric groups relative to Qwen2-VL-7B-Chat. Fine-grained accuracy decreases by 2.4pp, from 87.9% to 85.5%, and the general benchmark average decreases by 1.3pp, from 62.4% to 61.1%. As shown in Figure 8, this is the smallest fine-grained change among the major ablation categories, indicating that instruction-finetuning data contributes less to fine-grained knowledge than base model selection and pretraining strategy in our comparisons.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 summarize how each ablated factor narrows the gap between weaker and stronger VLMs on fine-grained and general VQA metrics. Starting from the LLaVA architecture (Liu et al., 2023) with Vicuna-7B (Chiang et al., 2023) and CLIP ViT-L/14 (Radford et al., 2021), the combined changes raise fine-grained performance from 52.8% to 73.4% and general VQA performance from 39.3% to 49.9%. The decomposition shows two different empirical roles: switching to DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining disproportionately improve fine-grained accuracy, whereas switching the LLM to Qwen2-7B (Yang et al., 2024) explains most of the general VQA increase.

Even after these substantial gains, Table 1 leaves a remaining 12-point fine-grained gap between our ablated model and Qwen2-VL-Chat. We consider two plausible explanations for this residual margin: architectural differences and pretraining data scale.

Architectural differences may contribute, but the observational results in Section 3.4 and the training ablations in Section 4.3 point most strongly to pretraining scale. Our LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) pretraining runs each use fewer than 1M images and captions, corresponding to roughly 200M and 400M tokens, respectively. By comparison, Wang et al. (2024b) report pretraining Qwen2-VL on 1.4T tokens, which is orders of magnitude larger than our ablation scale. This scale difference provides the most plausible explanation for the remaining fine-grained margin and reinforces the quantitative importance of extensive pretraining for high-accuracy vision-centric VLMs.

5 LIMITATIONS & CONCLUSION

Our study provides controlled evidence about which model and training choices move fine-grained accuracy, but the scale of our ablations also defines the main limitation. Due to computational constraints, our training comparisons use <1M data points rather than the billion-scale training used by some newer VLMs, so the exact margins we report may change under much larger pretraining regimes. Newer pretraining strategies may also interact differently with fine-grained visual understanding, and evaluating those interactions at scale is an important direction for future work.

In this work, we systematically evaluate state-of-the-art vision-language models (VLMs) on fine-grained classification benchmarks and show that these accuracies reveal capability differences not captured by standard general VQA averages. Across comparisons with encoder baselines and 22 controlled ablations, the evidence shows that stronger LLMs improve both metric groups, while stronger vision encoders, pretraining, and LLM-unfrozen pretraining are especially important for closing fine-grained accuracy gaps. These findings clarify which existing design choices most reliably improve vision-centric performance and support more practical VLM deployment in settings that require precise visual recognition.

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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: **Detailed aggregate results of model ablations.** Each row fixes the listed vision encoder, LLM, data sources, and trainable weights, enabling direct comparison of how component and training choices change average fine-grained and general VQA scores.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMU	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: **Detailed general VQA results for model ablations.** Scores are reported separately for the general VQA benchmarks used to form the ablation average, making the metric-level consistency of each comparison visible.

A FULL RESULTS

We provide the full numerical ablation record in three complementary tables: aggregated performance in Table 2, general VQA benchmark scores in Table 3, and fine-grained classification scores in Table 4. Together, these tables expose the exact metric-level comparisons behind the summary margins discussed in the main text.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: **Detailed fine-grained classification results for model ablations.** Scores are reported for ImageNet, Flowers, Pets, and Food so that fine-grained gains can be compared across datasets rather than only through the aggregate average.

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

As described in Section 3.2, we evaluate both VLMs and their base vision encoders on four traditional classification benchmarks so that VLM accuracies can be compared directly with encoder-only baselines:

ImageNet-1K (Deng et al., 2009): A foundational image classification dataset organized around the WordNet hierarchy, with broad supercategories that contain multiple fine-grained subcategories. The ImageNet test set contains 50,000 images distributed across 1,000 classes, making it the largest-class-count benchmark in our fine-grained evaluation suite.

Because of the ImageNet/WordNet hierarchy, some classes are too visually distinct from the rest of the label space to support hard multiple-choice distractors; for example, only three classes fall under the “person” category. To preserve difficulty while keeping the evaluation tractable, we use the human-annotated incorrect model predictions from Recht et al. (2019) to select images for which hard negative answer choices can be constructed. From 19,056 annotated examples, this filtering yields 5,457 images covering 928 of the original 1,000 classes. Because the resulting class distribution is uneven, we report mean-per-class accuracy for ImageNet.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): A 102-class flower species dataset with high intra-class variation and strong inter-class similarity. The Flowers test set contributes 6,149 examples to the fine-grained evaluation.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): A 37-class pet category dataset with variation in pose, lighting, and occlusion. The Pets test set contributes 3,669 examples.

Food-101 (Bossard et al., 2014): A 101-class food dataset with substantial visual diversity within each class. The Food test set contributes 25,250 examples, providing the largest number of images among the four benchmarks.

Question generation. For all fine-grained classification test sets except ImageNet, we generate hard negatives by running zero-shot classification with an OpenCLIP ViT-L/14 trained on LAION (Schuhmann et al., 2022), following Geigle et al. (2024). For each image, the top four incorrect

predictions become distractors and are shuffled with the correct label to produce the five answer choices.

Prompt formatting. In the multiple-choice VQA format, each classification query asks “What type of object is in this photo?” The answer choices are presented as “A. Option 1\n B. Option 2\n C. Option 3\n D. Option 4\n E. Option 5”. Because VLMs are trained with different multiple-choice prompt conventions, we use each model’s VLMEvalKit format (Duan et al., 2024); for example, some prompts include “Answer with the option’s letter from the given choices directly.”

For CLIP models, we use standard zero-shot classification rather than a question format. The class names passed to the text encoder use the template “a photo of a {classname}”, providing the encoder-only baseline for the same label set.

B.2 GENERAL VQA

For the initial observational comparison in Section 3, we use the scores reported on the OpenVLM Leaderboard (Duan et al., 2024). The default leaderboard average aggregates eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). This reported average is used only for Figure 2, where it provides the general VQA axis for comparing against fine-grained accuracy.

For all other evaluations, including existing models and our trained ablation models from Section 4, we use a smaller set of multiple-choice VLM benchmarks to make the comparison format closer to our fine-grained evaluation. Specifically, we report three benchmarks:

MMBench (Liu et al., 2024b): A 2,948-example benchmark covering a wide range of VLM capabilities across perception and reasoning. Although some examples are labeled as “fine-grained perception”, their categories, such as action recognition, attribute recognition, and OCR, differ from the object-category fine-grained classification datasets used in our main evaluation.

MMMU (Val) (Yue et al., 2024): A large-scale multimodal benchmark covering a broad distribution of domains and applications. Although the test set contains 10,500 examples, most works report performance on the 900-question validation split. MMMU measures general knowledge grounded in visual perception across art, science, engineering, medicine, business, and humanities.

MMStar (Chen et al., 2024b): A 1,500-question benchmark selected by human annotators. The examples cover diverse domains and low-level model capabilities, making MMStar a complementary general VQA metric.

Evaluation procedure. For general VQA benchmarks, we use the VLMEvalKit evaluation code without modification (Duan et al., 2024). Existing models use their predefined evaluation procedures, and our trained models use the LLaVA evaluation code because all of them are finetuned on LLaVA-Instruct data.

C TRAINING

We use the LLaVA-1.5 training codebase (Liu et al., 2023) as the common implementation framework for the ablations. For most experiments, we keep the LLaVA architecture and training procedure fixed so that metric differences reflect the factor being compared.

Pretraining. Each pretraining run uses one epoch over either LLaVA pretraining data from CC-3M images and captions or Molmo pretraining data from PixMo-Cap. For PixMo-Cap, we use only the “caption” field and exclude the raw audio transcript. The pretraining batch size is 256. In the connector-only condition, only the randomly initialized MLP connector is tuned, using a learning rate of 1e-3. For the pretraining-strategy ablation, the connector is trained alone for the first 20% of steps, after which the LLM is unfrozen and trained with a learning rate of 2e-5 for the remaining 80% of steps.

Finetuning. For finetuning, every model is trained for one epoch on the LLaVA-1.5 finetuning data, which combines LLaVA-Instruct with VQA training data from multiple datasets to support benchmark evaluation. During finetuning, the LLM learning rate is 2e-5 and the batch size is 128.

The vision encoder remains frozen during both pretraining and finetuning, keeping encoder weights fixed across comparisons.

For the Qwen2-VL architecture ablation, we start from Qwen2-VL-7B Base and apply the same finetuning method while unfreezing only the LLM.

Compute. All models are trained on 4 nodes with 4 A100 (40GB) GPUs per node. Training typically finishes within 20 hours, with runtime varying according to CLIP model size.

LLaVA vs. PixMo data. The main text compares LLaVA and PixMo pretraining data as two caption-quality conditions. LLaVA pretraining data comes from CC-3M and uses web-scraped captions that are commonly lower in quality and relevance. PixMo-Cap is collected from human annotators who record a 60–90 second audio description of an image, which is then converted into an image caption (Deitke et al., 2024). A surface-level length comparison highlights the contrast: LLaVA captions average 9.8 words, whereas PixMo captions average 169 words.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”